

DIET AND DISEASE — THE ISRAELI PARADOX: POSSIBLE DANGERS OF A HIGH OMEGA-6 POLYUNSATURATED FATTY ACID DIET

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ABSTRACT. Israel has one of the highest dietary polyunsaturated/saturated fat ratios in the world; the consumption of omega-6 polyunsaturated fatty acids (PUFA) is about 8% higher than in the USA, and 10–12% higher than in most European countries. In fact, Israeli Jews may be regarded as a population-based dietary experiment of the effect of a high omega-6 PUFA diet, a diet that until recently was widely recommended. Despite such national habits, there is paradoxically a high prevalence of cardiovascular diseases, hypertension, non-insulin-dependent diabetes mellitus and obesity — all diseases that are associated with hyperinsulinemia (HI) and insulin resistance (IR), and grouped together as the insulin resistance syndrome or syndrome X. There is also an increased cancer incidence and mortality rate, especially in women, compared with western countries. Studies suggest that high omega-6 linoleic acid consumption might aggravate HI and IR, in addition to being a substrate for lipid peroxidation and free radical formation. Thus, rather than being beneficial, high omega-6 PUFA diets may have some long-term side effects, within the cluster of hyperinsulinemia, atherosclerosis and tumorigenesis.

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HIGH CONSUMPTION OF LINOLEIC ACID IN ISRAEL

Linoleic acid (LA) is an essential fatty acid whose minimum requirement is estimated at 1% of the energy intake (en%) (1). It is so far the most abundant polyunsaturated fatty acid (PUFA) in western diets (2). For all practical purposes the “P” in the polyunsaturated/saturated (P/S) ratio stands for LA. LA consumption in Israel is very high, and according to a 1976 estimate the P/S ratio is 1.6 (3), far higher than in other countries: 0.63 in the USA (4), 0.7 in Belgium, 0.6 in The Netherlands, 0.4 in Sweden and Italy, and 0.3 in Poland (5).

Surveys of urban families (6–8) point to a low intake of omega-3 PUFA, and LA intake of about 10

en%, whereas calculations based on recent food balance sheets lead to even higher values (Central Bureau of Statistics 1985/86). The most convincing evidence for the extremely high LA consumption comes from the fatty acid analyses of subcutaneous adipose tissue, which reflect the intake of essential PUFA over a period of 1–3 years (9). Five different studies (4,10–13) in which the fatty acid composition of adipose tissue fat was determined in Israelis yielded mean LA contents of total fatty acids ranging from 22.1% in 1986 (4) to 27.2% in 1994 (13). If 24% is taken as an average figure, the daily intake, calculated by the formula proposed by Beynen et al. (9), is 31.8 g, which would represent 11.8 en% for a caloric intake of 2,500 Cal. In the USA, where the intake of LA has been increasing progressively over the past decades (14), the LA content of adipose tissue has reached 16% (12,15), compared with levels of <10% in most Northern European countries (5,16), 12–14% in Germany, Italy and The Netherlands (5,16,17), and 16.7% in Belgium (5). The reason why

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Israeli Jews, irrespective of ethnicity or religious observance, consume a large amount of LA is that in this country the marketed oils and the margarines made from them are rich in LA (Central Bureau of Statistics 1985/86). Soybean oil is the major oil consumed. Other vegetable oils are cottonseed, corn, sunflower and safflower, whereas olive oil, which has <10% LA, is not consumed in large amounts. Total intake of fat of vegetable origin in Israel is almost double that of animal fat, the latter originating from fatty meat and dairy products. In most western countries, the ratio is reversed, animal fats being consumed in much greater amounts than vegetable oils.

CANCER AND CARDIOVASCULAR DISEASES, HYPERTENSION, OBESITY AND NIDDM IN ISRAEL

A statistically high prevalence of diseases considered to be associated with hyperinsulinemia (HI) has been observed in the four main ethnic groups originating from Yemen, the Middle East, North Africa and Europe-America, which constitute the adult Israeli Jewish population. According to data compiled by the Central Bureau of Statistics and the Israel Ministry of Health (Table 1), cardiovascular disease (CVD) and cancer mortality rates — selected by sex in Israeli Jews — rank high on the international mortality list, especially for women, despite a mean plasma cholesterol concentration of <5.2 mmol/l (210 mg/dl) (18–20).

Because the treatment of many cancers and cardiovascular diseases has greatly improved, it is essential to use incidence data to assess the impact of prophylactic measures on the one hand, and environmental factors, including diets, on the other. These factors are also reflected in migrant populations 10 to 20 years after moving to a new country, and in adults as well as children who relocate. Data presented in Table 2 show a substantial increase in the incidence of cancers generally accepted to be related to fat consumption in all ethnic groups resident in Israel. (Similar data on heart-associated diseases are not available.) In addition, the incidence of diabetes mellitus in Jews of all ethnic groups is high (Table 1), and generally increases with length of residence in Israel (21–23).

Furthermore, the Jewish population, possibly influenced by genetic and lifestyle factors (24), appears to be more obese than the American and considerably more than the European populations (22,25,26) despite consuming less calories and fats (Table 3). Even more remarkable are the highly

Table 1. Deaths caused by diabetes mellitus, cardiovascular and malignant diseases in Jews, and non-Jews, 1987–89. Standardized rates per 100,000 population aged 25+

	Male		Female	
	Jewish	Non-Jewish	Jewish	Non-Jewish
Diabetes mellitus	15.3	9.7	20.3	13.3
Heart and circulatory				
Hypertensive	6.7	4.0	10.0	6.0
Acute myocardial	110.3	39.7	81.7	30.0
Other heart ischemia	67.7	21.3	49.0	16.7
Other heart	70.7	36.3	73.3	44.0
Cerebrovascular	60.7	27.0	68.0	37.3
Other circulatory	6.3	2.3	5.7	3.0
Total heart and circulatory	322.4	130.6	287.7	137.0
Malignant diseases				
Stomach	10.7	2.3	6.3	3.7
Colon	15.3	2.0	14.7	2.0
Rectum	4.0	0.3	3.7	1.0
Trachea, bronchus, lung	27.0	14.3	11.0	2.7
Breast	29.0	6.0		
Cervix, uterus	1.7	1.0		
Leukemia	7.3	3.3	6.0	3.7
Other	80.0	23.7	65.0	17.0
Unspecified	6.0	2.0	3.3	1.3
Total neoplasms	150.3	47.9	140.7	38.4

Source: Central Bureau of Statistics, Ministry of Health, Special series No. 923, Jerusalem, 1992.

significant differences in heart, circulatory, and cancer disease mortality rates between Israeli Jews as compared with non-Jewish citizens (mostly Arabs and Druze), even when living in the same district. The latter consume a diet rich in olive oil (Table 1).

LA INVOLVEMENT IN DELETERIOUS METABOLIC PATHWAYS

Dietary PUFA have long been known as cholesterol-lowering fatty acids (27,28). The most desirable level for LA intake has not been determined. Most investigators believe that an intake of 6–10 en% is optimal (29,30). However, in spite of widespread enthusiasm for increasing PUFA in the diet, some investigators have voiced their concern. This is because recent studies from a variety of disciplines suggest possible deleterious effects of a high consumption of LA, including the following:

- Interfering with the metabolism of alpha-linolenic acid omega-3 (31), another essential fatty acid whose metabolites play an important role in the normal development of the retina, brain and clotting mechanism (32).

Table 2. Mean incidence of cancers related to fat consumption. Rates per 1000,000 in 1960, 1989, and ratio between them

	Israeli-born			Migrants Europe & America			Migrants Africa & Asia			Non Jews		
	1960	1989	89/90	1960	1989	89/90	1960	1989	89/90	1960	1989	89/90
Females												
Colon	7.5	24.4	3.25	14.2	23.8	1.67	5.2	12.2	2.35	2.3	4.7	2.04
Rectum	2.3	12.6	5.47	8.8	14.2	1.61	2.7	11.6	4.30	1.6	2.9	1.81
Breast	35.9	104.0	2.90	55.5	87.0	1.57	22.2	57.2	2.50	8.1	21.2	2.61
Corpus uteri	7.9	14.3	1.81	10.7	11.1	1.04	4.6	7.1	1.54	2.6	6.8	2.61
Ovary	7.3	17.9	2.45	15.3	14.4	0.94	4.3	7.9	1.83	2.7	4.5	1.66
Males												
Colon	9.3	19.9	2.14	12.2	29.0	2.38	4.9	15.0	3.06	4.5	5.1	1.13
Rectum	6.7	7.4	1.10	9.7	17.8	1.83	3.1	12.4	4.00	2.0	5.7	2.85
Lung	17.3	31.1	1.80	29.6	28.5	0.96	22.7	31.3	1.38	24.6	24.4	0.99
Prostate	10.8	27.1	2.51	13.2	20.7	1.57	11.4	20.5	1.80	3.1	13.1	4.22

Data on female lung cancer not shown because of significantly different smoking habits between Jews and non-Jews.

Sources — Israel Cancer Registry, Cancer in Israel, Facts and Figures, 1989; Israel Ministry of Health, Department of Epidemiology, Jerusalem, 1993.

Table 3. Mean food intake and composition in Israel, USA^a, UK and Greece^b

	Israel	USA	UK	Greece
Total calories (TC)	2.999	3.455	3.040	3.557
Protein % of TC	14.0	12.1	11.7	12.1
Fat % of TC	32.7	43.9	42.2	36.4
Carbohydrate % of TC	53.3	44.0	46.1	51.5

^aFAO Food Balance Sheets 1979–1981.

^bWorld Health Statistics Annual, World Health Organization, Geneva, 1984.

- Augmenting the ratio of linoleic to alpha-linolenic acid, and thus affecting a variety of pathophysiologic processes, ranging from immune-inflammatory reactions to atherogenesis and tumorigenesis (33).
- Increasing the susceptibility of low density lipoprotein (LDL) to oxidative modifications (34,35), and with it the risk for acute myocardial infarction and coronary thrombosis (36). Furthermore, increasing the amounts of LA and LA hydroperoxide could reduce the activity of prostacyclins in the vascular wall (37). Also, there are effects on the production of thromboxane B₂ and, thereby, the tendency for platelet aggregation (see Fig. 1).
- Reducing the level of high density lipoprotein (HDL) cholesterol (36,38), and augmenting the risk for coronary heart disease (CHD) mortality (19,20). Lipid peroxidation mediated by free rad-

icals and/or hydroxy radicals is considered to be associated with the activation of radical scavengers, initiation and development of atherosclerosis (39), and decreasing the content of antioxidant vitamin E (40).

- Increasing the secretion of insulin, and/or reducing insulin catabolism (41,42), causing impaired insulin action (43,44) and leading progressively to insulin resistance (IR) (45,46), the negative effects of which have been grouped together as the insulin resistance syndrome (IRS) or syndrome X (45). In this context it is important to note that insulin activates phospholipase A₂, and consequently augments eicosanoid formation and substrate for lipid peroxidation (47,48). The roles of free radicals and lipid peroxidation in cancer etiology (49) and atherosclerotic processes (39) have been documented extensively. The same relationships hold for eicosanoids formed from dietary omega-6 lipids affecting thrombosis, vasospasm, arrhythmia, and chronic inflammatory processes (50).

HYPERINSULINEMIA INVOLVEMENT IN CVD, ATHEROGENIC PROCESSES, DYSLIPIDEMIA, HYPERTENSION AND OBESITY

Epidemiologic studies show that HI, both fasting and postprandial, increases the risk factor for CVD in nondiabetic (51) and NIDDM subjects (51,52), and has an important role in atherogenic processes

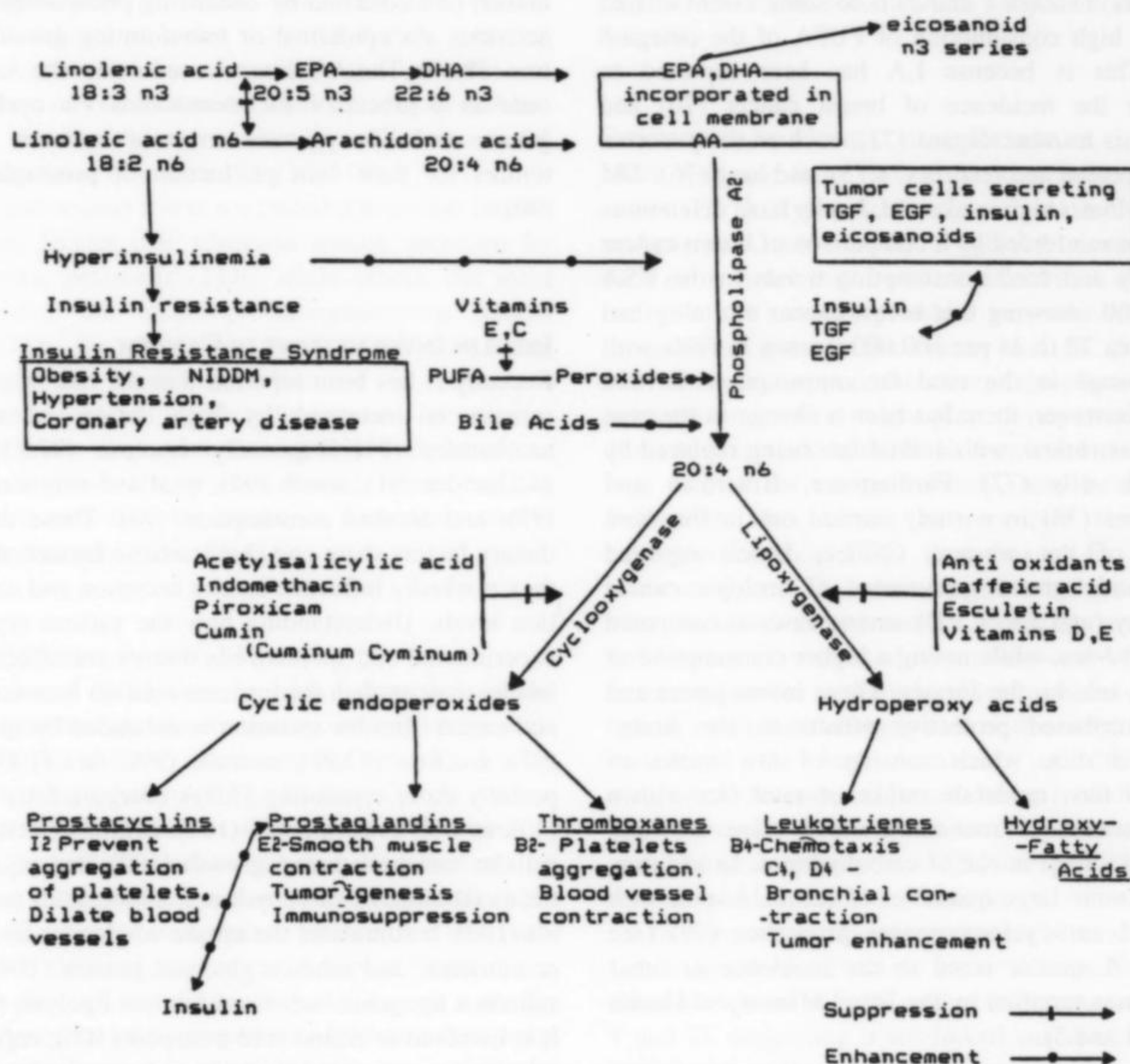


Fig. 1. Polyunsaturated fatty acids (PUFA), insulin, eicosanoid, cardiovascular disease and tumor inter-relationships. EPA = eicosapentaenoic acid; DHA = docosahexaenoic acid; AA = arachidonic acid; TGF = transforming growth factor; EGF = epidermal growth factor.

(45,46,50-53), while experimental data reviewed by Stout (54) attribute to insulin the following atherogenic properties: a) increasing formation and decreased regression of lipid plaques, b) inducing proliferation of smooth muscle cells, c) stimulating the synthesis of connective tissue, d) enhancing cholesterol synthesis and increasing LDL receptor activity, and e) stimulating growth factor activities.

In addition, HI and IR exert effects on lipid metabolism, causing hypertriglyceridemia and a decrease in HDL cholesterol (45,46,55-58), and in some cases hypercholesterolemia (56,59). Both conditions might lead to hypertension (55,60-62). IR is a feature of NIDDM (63) and represents the hallmark

of obesity (29,30,64-67). It is also associated with desaturase deficiency in membrane phospholipids, and consequently a high LA and low arachidonic acid (AA) content (68,69).

LA, EICOSANOIDS, AND CANCER INCIDENCE AND PROCESSES

The relationship between diet and cancer is a notorious source of both frustration and excitement. However, in light of the above, one may ask whether the high incidence of malignancies among Israeli Jews (especially when compared with Israeli

non-Jews) (Tables 1 and 2) is to some extent related to their high consumption of PUFA of the omega-6 type. This is because LA has been reported to increase the incidence of breast cancer (70) and metastasis to other organs (71), such as the pancreas (72,73), colon and rectum (74,75), and lung (76). The concept that a high intake of LA may have deleterious effects is reinforced by a comparison of breast cancer mortality and food consumption trends in the USA since 1960, showing that breast cancer mortality had risen from 22 to 34 per 100,000 women in 1990, with little change in the total fat consumption. In that period, however, there has been a change in the type of fats consumed, with animal fats being replaced by vegetable oils (77). Furthermore, Bitterman and colleagues (78) in a study carried out in the Acre district of the western Galilee, Israel, reported significantly greater prevalence of urologic cancer morbidity (and also CVD) among Jews as compared with non-Jews, while noting a higher consumption of high LA oils by the former. These investigators and others attributed protective effects to the Arabs' traditional diet, which consists of low intake of saturated fats, moderate intake of total fats with a preponderance of monounsaturated olive oil fatty acid, and a high intake of carbohydrates. In addition, they consume large quantities of cumin, a spice with attributed anti-cyclooxygenase properties (79) (see Fig. 1). A similar trend in the incidence of other cancers was reported by the Israel Ministry of Health (Tables 1 and 2).

Some reports suggest that LA, the parent fatty acid of the omega-6 family of fatty acids, potentiates tumorigenesis by providing structurally and functionally essential fatty acids to dividing cells, and by serving as precursor to eicosanoid metabolites of AA.

Leukotrienes C_4 and D_4 might act as tumor-enhancing agents (80,81), and prostaglandin E_2 (PGE_2) in turn may be involved in tumor development, promotion and immunosuppression (13,81,82), reducing macrophage tumoricidal activities and inhibiting interleukin 2 production, which activates the natural killer cell cytotoxic T-cell activity (Fig. 1). Furthermore, increased levels of PGE_2 have been associated with aggressive growth patterns of both basal cell and squamous cell skin carcinoma in humans (83-85). On the other hand, factors inhibiting the cyclooxygenase enzyme system, such as indomethacin, piroxicam, or the spice cumin (79) (Fig. 1), have been reported to reduce the incidence of colon (86) and possible urinary tract cancer (78,87). Similarly, blocking the lipoxygenase pathway by esculetin suppresses PC3 prostate cancer cells (86) or mastocytoma cells in culture (80). Certain tumors also augment the for-

mation of eicosanoid by enhancing phospholipase A_2 activities via epidermal or transforming growth factors (88,89). The AA thus released enters the AA cascade as a precursor for eicosanoids via cyclooxygenase and lipoxygenase enzymes systems, or to tumors for their own production of prostaglandins (90).

INSULIN INVOLVEMENT IN CANCER

Recently it has been reported that the risk of certain cancers is increased by high intake of monosaccharides (91), especially fructose (92,93), disaccharides (91), starch (94), total and saturated fats (95), and alcohol consumption (96). These diverse dietary factors share one characteristic feature; that is, they markedly influence insulin secretion and circulation levels. Understanding how the various types of experimental and spontaneous tumors are affected by insulin may explain the intricate relation between diet and cancer. Insulin secretion is enhanced by glucose (97), fructose (92,98), sucrose (99), fats (100), especially those containing PUFA omega-6 fatty acids (40), as well as by alcohol (101). Insulin affects both cellular metabolism and growth (102), and regulates the metabolism of carbohydrates, amino acids and lipids (103). It stimulates the uptake of glucose and other nutrients, and inhibits gluconeogenesis (104). Insulin is a lipogenic factor and inhibits lipolysis (105). It is involved in amino acid transport (106), regulates cellular gene expression (107) and protein synthesis (108), and augments eicosanoid formation as described above.

Many reports have emphasized the involvement of insulin in various processes associated with tumor cell proliferation such as DNA synthesis (109), and cell mitosis and growth (110). These results are reinforced by studies reporting high insulin activity in several tumors (13,111), or elevated plasma concentrations of insulin-like substances, sometimes associated with IR in tumor-bearing animals or patients (13,91,112-116). In ovarian and endometrial cancer patients, HI has been reported to be accompanied by hyperlipidemia and hypoglycemia. Most importantly, all blood parameters appear to return to normal after surgical intervention in non-obese women (13). Insulin resistance is absent in tumor cells (117), conferring on them serious advantages, especially in hypoglycemic states. The role of insulin in cancer promotion was reviewed by Yam (118), and is illustrated by the results of studies with alloxan diabetic animals and other models (114,115,119).

Noteworthy are the facts that: a) certain tumors

grow more slowly in those animals, b) they had a lower incidence of tumor than non-diabetic ones, c) insulin administration enhanced the growth and incidence of certain tumors, and d) alloxan-diabetic mice became hyperinsulinemic after successful transplantation of tumors (92,114–116).

In this respect it was suggested that certain tumors produce factors that stimulate insulin secretion by pancreatic beta-cells (114), while others, the most aggressive, are insulin-producing/secreting tumors (115,116). On the other hand, very convincing evidence for insulin involvement in cancer is provided by agents which reduce insulin secretion or action such as somatostatin (120) and interferon (121), or block the tyrosine kinase receptor (122). These factors may be inhibitors for certain cancer cells and tumors (121,123,124).

THE OPTIMAL N-6/N-3 RATIO

The HI and IR induced by high LA intake may be prevented by fish oils containing omega-3 fatty acids (42). In this context, various studies provide evidence linking a high dietary omega-6/omega-3 ratio with an increased risk of diabetes, CVD, and a wide variety of pathophysiologic mechanisms and complications including insulin action and secretion, and/or IR (125–131). The known competition between the two pathways of essential fatty acids is very difficult to untangle biologically. Thus, it is not known with any confidence in what proportions C20 and C22 long-chain fatty acids will be produced from a given mixture of linoleic and alpha-linolenic acids fed to any animal (132). Further epidemiologic evidence is provided by Raheja and colleagues (129). They reported a sharp increase in the prevalence of NIDDM and coronary artery disease (CAD) in the upper socioeconomic classes in India after they adopted diets high in total fat, with high concentrations of omega-6 fat and a high omega-6/omega-3 ratio in lipids. Omega-3 fatty acids may be able to alleviate many of the metabolic abnormalities associated with obesity (130), hypertension (15) and hypertriglyceridemia (126). Furthermore, a low incidence of breast cancer has been observed in Greenland Inuits despite their relatively high fat diet, and also in Japanese women who conserve their traditional diet which contains fats derived from marine sources that are rich in omega-3 PUFA. Thus, the ratio of the two fatty acids in the diet is important and would seem to be on the increase when compared to the diet in the past. Mesolithic man had a diet ratio of n-6:n-3 of 1–4:1 (131), the European diet reaches 10–14:1 (67), and the Israeli present average ratio is approximately 22–

26:1, as shown in subcutaneous fatty acid composition (13).

SUMMARY AND CONCLUDING REMARKS

Much evidence links HI and IR to atherogenic processes, CVD, dyslipidemia, hypertension, NIDDM, malignancies, and obesity. High LA consumption such as in Israel might exert a deleterious effect directly, or via enhancing insulin secretion effects, or by inducing glucose intolerance and IR (35,41,42, 129). One may suppose that every factor having a hyperinsulinemic effect might play an important role in diseases associated with HI. However, LA, the parent fatty acid of the omega-6 family, may be more effective by also serving as precursor for certain eicosanoids such as PGE₂, thromboxane B₂, and leukotrienes C₂ and D₂ (whose negative influences have been discussed above), and by providing structurally and functionally essential fatty acids to dividing cells in tumors, and/or perhaps in those involved in atherogenic processes. Looking at it in this way, one may understand the high incidence of the above-mentioned diseases in Israeli Jews, including their higher prevalence of obesity as compared with U.S. and European populations (23), despite their consumption of less calories (Table 3).

In comparison to western women, it seems that the Israeli female is relatively more susceptible to diseases associated with HI, and to malignancies (Tables 1 and 2), suggesting a sex-linked susceptibility. The fact that diseases associated with HI increase among Jewish immigrants with length of residence in Israel also reinforces the assumption of dietary or other environmental causes. Further possible dietary effects on cancer through influencing insulin metabolism have been hypothesized by Yam et al. (115). It was suggested that an elevated insulin status, induced by diet or other factors, might activate dormant cells, initiate the development of preneoplastic lesions, or stimulate the growth of existing tumors. The incidence of cancer, CVD and associated diseases, and obesity in human populations may be diminished by dietary restriction aimed at preventing a hyperinsulinemic state. This includes not only such well-known factors as reduced calorie and fat intakes, but also the avoidance of gorging, and perhaps a change in the number and size of meals eaten.

It is also recommended, according to an almost universal consensus, that active subjects who are in energy balance, may consume up to 30% of their total energy intake from dietary fat: 7% of energy from saturated fats, 10–15% from monounsaturated fatty acids, and up to 10% from polyunsaturated fatty acids

(30,133). It is suggested that the intake of hydrogenated oils and their products should also be reduced. The optimal ratio of omega-6 to omega-3 fatty acids has yet to be decided, and there are reasons to counter the trend for its increase (67).

An inverse association with antioxidants has been noted for several cancers and, according to the Food and Agricultural Organization (FAO), subjects whose diets have a P/S ratio of up to 0.45 are recommended to consume antioxidants such as vitamin E (4 mg/day for men, 3 mg/day for women), selenium (75 g/day for men, 60 g/day for women), vitamin C (40 mg/day), and beta carotene. Israeli Jews possibly require higher amounts since their dietary and adipose tissue P/S ratio is higher.

Moderate exercise training can enhance insulin sensitivity, lower plasma triglyceride and insulin levels, lower blood pressure, and increase HDL cholesterol concentrations (134,135). In addition, early screening for insulin status may help identify people at risk. Insulin levels should be investigated in patients in whom cancer has been diagnosed or is suspected.

Attention should be focused on dietary factors inhibiting eicosanoid enzyme systems, such as cumin, or other modalities able to reduce HI, and also on the influence of different types of fatty acids on the biology of human cancer. Special consideration should be given to the dietary balance between n-6 and n-3 fatty acids, and studies on LA metabolism, especially in women.

Diets high in LA were highly recommended in the past; however, Israel, which has the highest intake, did not appear to show benefits in heart disease mortality. This constitutes the Israeli paradox and was alluded to in 1976 (10). This review summarizes the biologic and epidemiologic considerations underlying the paradox, which suggest that a high PUFA diet (in particular one rich in LA) may not be beneficial for the population at large.

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